3.2 Multiplication of Algebraic Expressions

While doing the product of algebraic expressions, we should follow the steps given below.

Step 1: Multiply the signs of the terms. That is, the product of two like signs are positive and the product of two unlike signs are negative.

Like signs	$(+)\times(+)=+$	$(-)\times(-)=+$
Unlike signs	$(+)\times(-)=-$	$(-)\times(+)=-$

Step 2: Multiply the corresponding co-efficients of the terms.

Step 3: Multiply the variable factors by using laws of exponents.

If 'x' is a variable and m, n are positive integers then,

$$x^m \times x^n = x^{m+n}$$

For example, $x^3 \times x^4 = x^{3+4} = x^7$

Think

Every algebraic expression is a polynomial. Is this statement true? Why?

DO YOU KNOW?

A polynomial is a special kind of algebraic expression. The difference between an algebraic expression and a polynomial is,

Algebraic Expression

May contains whole numbers, fractions, negative numbers as the power of their variables.

Example: $4x^{3/2} - 3x + 9$

$2y^2 + \frac{5}{y} - 3$, $3x^2 - 4x + 1$

Polynomial

contains only whole numbers as the power of their variables.

Example: $4x^2 - 3x + 9$

$2y^6 + 5y^3 - 3$

Note

Product of two terms is represented by the symbols ( ), dot (.) or $\times$.

For example,

multiplying $4x^2$ and xy can be written in any one of the following ways.

$(4x^2)(xy)$	$4x^2 \times xy$	$4x^2(xy)$	$(4x^2)\times xy$	$4x^2 \cdot xy$

3.2.1 Multiplication of two or more monomials

Consider that, Geetha buys 3 pens each @ ₹5, how much she has to pay to the shopkeeper?



Geetha has to pay to the shopkeeper $= 3 \times ₹5$

$= ₹15$

8th_Maths_English Medium_Chapter 3.indd 77 23/11/2022 15:40:01